

Retake Exam

Duration 1h30

All answers must be rigorously justified. Efforts should be made to ensure clear and well-structured presentation. No reference materials or calculators are allowed.

Exercise 1. (7 points)

I/ Let f be the function defined by $f(x) = \sqrt{4 + 3x}$.

- 1. Determine the fixed points of the function f .*
- 2. Study the nature of the sequence (u_n) defined by:*

$$\begin{cases} u_0 \geq 0, \\ u_{n+1} = f(u_n), \quad \forall n \in \mathbb{N}. \end{cases}$$

II/ Study the nature of the sequence defined by:

$$v_n = \frac{2^{2n+3} + 3^n n + (-1)^{n-1} + E(n)}{4^n}.$$

Exercise 2. (3 points)

Let I be a non-empty interval in $\mathbb{R}$, and let $f : I \rightarrow \mathbb{R}$ be a continuous function. Show, using the Intermediate Value Theorem, that if $(f(x))^2 = 1$ for all $x \in I$, then f is constant.

Exercise 3. (5 points)

Let $(S_n)_{n \geq 1}$ be the sequence defined by $S_n = 1 + \frac{1}{2^3} + \frac{1}{3^3} + \cdots + \frac{1}{n^3}$.

- 1. Let f be the function defined by $f(x) = \frac{1}{x^2}$. Show that for all $k \in \mathbb{N}^*$, we have:*

$$\frac{2}{(k+1)^3} < f(k) - f(k+1) < \frac{2}{k^3}.$$

- 2. Show that the sequence (S_n) is bounded above.*
- 3. Determine the nature of the sequence (S_n) .*

Exercise 4. (5 points)

Let f be the function defined by $f(x) = \arctan\left(\frac{\sqrt{1-x^2}}{x}\right)$.

- 1. Determine the domain of definition of f .*
- 2. Can f be extended continuously at $x = 0$?*
- 3. Determine the set on which f is differentiable, and compute its derivative.*